Git and GitHub

- Learn the Basics
  - What is Version Control?
  - Why use Version Control?
  - Git vs Other VCS
  - Installing Git Locally
- What is a Repository
  - git init
  - git config
  - Local vs Global Config
  - Repository Initialization
  - Working Directory
  - Staging Area
  - Committing Changes
  - Intro and Git Commands
  - .gitignore
  - Viewing Commit History
- Branching Basics
  - Creating Branch
  - Renaming Branch
  - Deleting Branch
  - Checkout Branch
  - Merging Basics
- Basic Collaboration
  - GitHub Essentials
    - Creating Account
    - GitHub Interface
    - Setting up Profile
      - Profile Readme
    - Creating Repositories
      - Private vs Public
  - Git Remotes
    - Cloning Repositories
    - Managing Remotes
    - Pushing / Pulling Changes
    - Fetch without Merge
  - Merge Strategies
    - Fast-Forward vs Non-FF
    - Rebase
    - Squash
    - Handling Conflicts
    - Cherry Picking Commits
- Collaboration on GitHub
  - Forking vs Cloning
  - Issues
  - Pull Requests
    - PR from a Fork
    - Collaborators
    - Creating PR
  - Labelling Issues / PRs
  - Saved Replies
  - Mentions
  - Reactions
  - Commenting
  - GitHub Discussions
- Best Practices
  - Commit Messages
  - Branch Naming
  - PR Guidelines
  - Code Reviews
  - Contribution Guidelines
  - Clean Git History
- Documentation
  - Markdown
  - Project Readme
  - GitHub Wikis
  - CITATION files
- Working in a Team
  - Collaborators / Members
  - GitHub Organizations
  - Teams within Organization
- GitHub Projects
  - Project Planning
  - Kanban Boards
  - Roadmaps
  - Automations
- Intermediate Git Topics
  - Git Stash Basics
  - History
    - Linear vs Non-Linear
    - HEAD
    - Detached HEAD
    - git log options
  - Undoing Changes
    - git revert
    - git reset
      - --soft
      - --hard
      - --mixed
  - Viewing Diffs
    - Between Commits
    - Between Branches
    - Staged Changes
    - Unstaged Changes
  - Rewriting History
    - git commit --amend
    - git rebase
    - git filter-branch
    - git push --force
  - Tagging
    - Managing Tags
    - Pushing Tags
    - Checkout Tags
    - GitHub Releases
  - Git hooks
    - What and Why?
    - Client vs Server Hooks
    - commit-msg
    - post-checkout
    - post-update
    - pre-commit
    - pre-push
    - Common Hooks
  - Git Patch
  - Submodules
    - What and Why use?
    - Adding / Updating
- GitHub Workflow
  - GitHub Actions
    - What are these?
    - Usecases
    - YAML Syntax
    - Workflow Triggers
    - Scheduled Workflows
    - Workflow Runners
    - Workflow Context
    - Secrets and Env Vars
    - Caching Dependencies
    - Storing Artifacts
    - Workflow Status
    - Marketplace Actions
  - GitHub CLI
    - Installation and Setup
    - Repository management
    - Issue Management
    - Pull Requests
    - Use in Automation
- Advanced Git Topics
  - Git Reflog
  - Git Bisect
  - Git Worktree
  - Git Attributes
  - Git LFS
- GitHub Developer Tools
  - GitHub API
    - REST API
    - GraphQL API
    - GitHub Apps
    - OAuth Apps
    - Creating Apps
  - Webhooks
- More GitHub Features
  - GitHub Gists
  - GitHub Packages
  - GitHub Codespaces
  - GitHub Security
  - GitHub Sponsors
  - GitHub Pages
    - Deploying Static Websites
    - Custom Domains
    - Static Site Generators
  - GitHub Copilot
  - GitHub Models
  - GitHub Marketplace
  - GitHub Education
    - Student Developer Pack
    - GitHub Classroom
    - Campus Program